Almanac 47 for Free42

SwissMicros DM42 / DM42n with the stock (Free42) firmware — user manual — version 2.3.0 (the programs of 2.1.0 Supercharger, your stack kept, UT / LT on +/-)

The full Almanac 47 of the C47 — the Sun, the Moon, the planets and the 58 navigation stars, 5 views and INFO — converted to Free42 3.3 with the DM42 graphics extension (the whole 400 × 240 screen). The menu, the screens and the values are the same as on the C47: in the Free42 core (SwissMicros source, release 3.3.10) each screen was compared pixel by pixel with the C47 version.

New in v2.3.0: the key **+/-** shows every time in UT or in **local time**, LT = UT + TZ (the variable TZ: your zone in hours, e.g. 3 STO "TZ" for UTC+3, -5 STO "TZ" for UTC−5). In LT the header shows the local date and time and LT, and rise, set, twilights and meridian passage are + TZ; GHA, Dec, Hc and Zn do not change, and DATE and UT are still asked in UT. The choice is the variable TZLT (0 UT, 1 LT), kept between runs; NAV creates TZ and TZLT with 0 when they are missing. NAVFULL: +/- on the menu or a view; NAVLITTLE: +/- redraws ALMANAC; MOON47: +/- for the times, 0 for the view from the south.

New in v2.2.1: NAVFULL, NAVLITTLE and MOON47 give you back your stack when they end: X, Y, Z, T and LASTX are as they were before XEQ "NAV" (or XEQ "MOON47"). Free42 has no STOS and no local registers, so the four levels and LASTX go into the variables S7X, S7Y, S7Z, S7T, S7L at the start and come back at the end (STO ST L, then the four RCL); the variables are deleted. With the 4-level stack of the HP-42S (Free42's dynamic stack keeps only those four). The screens and the keys are those of v2.1.0. On the C47 / R47 the same idea is the forum's STOS / RCLS trick.

It supports, and does not replace, the Nautical Almanac. Always cross-check the values.

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1. What you need

- A **DM42n** (recommended) or DM42 with the SwissMicros stock firmware (DM42 3.26 = Free42 3.3.10 or later). The DM42n has ample memory. For little memory there is NAVLITTLE: the ALMANAC screen with the Sun and the 58 stars only (section 10).
- Free42 on a PC or phone has only the 131 × 16 HP-42S screen: NAV runs there, but you see only the top left corner of each screen. Use the calculator for the views.

2. Files (build/free42)

File	What it is
NAVFULL.raw	the program NAV and its routines: all the views; the screen appears complete, at once, as on the C47
NAVINIT_FULL.raw	INIT for NAVFULL: the series matrices, valid 2000–2050
NAVINIT_FAST.raw	INIT for NAVFULL: fitted series for 2026–2030 (smaller, faster)
NAVLITTLE.raw	NAV for little memory: the ALMANAC screen, Sun and stars only (section 10)
NAVINIT_LITTLE.raw	INIT for NAVLITTLE: Sun, nutation and stars, valid 2000–2050
TBL_1.raw, TBL_5.raw	optional almanac tables (JPL) for 1 or 5 years from 1 Oct 2026 (section 11)
*.txt	the same programs as text (Free42 on a PC: Paste in PRGM mode)
MOON47.raw	the Moon phase on its own (section 12)

3. Loading and first start

- Connect the calculator by USB (it appears as a disk) and copy the .raw files into the PROGRAMS folder.
- The file name is the version: on the calculator the programs are always NAV and INIT.
- On the calculator: **SETUP → Load Program**, load NAVINIT_FULL.raw (or NAVINIT_FAST.raw).
- **XEQ "INIT"**: it builds the matrices and shows MATRICES READY. Then delete INIT (GTO "INIT", CLP): the matrices stay.

- Load NAVFULL.raw and **XEQ "NAV"**.

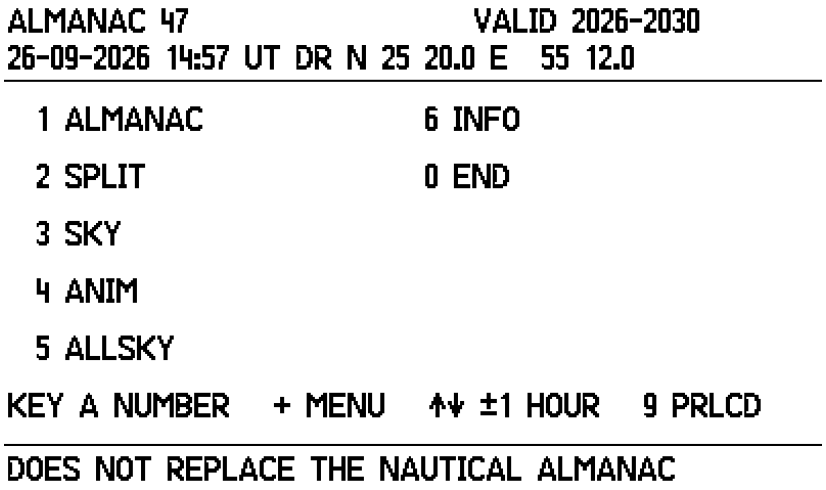
4. Starting NAV

NAV asks four numbers with INPUT (R/S keeps the value shown). Your stack is saved first; at each prompt the stack is clear and Y shows the format of that input (DATE Y.MMDD, UT H.MMSS, LAT D.MMm S -, LON D.MMm W -):

Input	Meaning	Example
DATE	UT date, YYYY.MMDD	2026.0926
UTC	UT, HH.MMSS	14.57
LAT	latitude DD.MMm, south negative	25.20 = 25° 20′ N
LON	longitude DDD.MMm, west negative	55.12 = 55° 12′ E

- **Time is UT (UT1)**: for a sight timed in UTC add DUT1 (under 0.9 s) — as in the almanac.
- **Signs:** latitude N +, S −; longitude E +, W −.

Then NAV switches to the 400 × 240 graphics mode (GrMod 3), computes the sky (the SINKING....ABOUT box shows meanwhile, only at the start) and draws the menu. 0 on the menu ends NAV and sets the normal screen again (GrMod 0). Under the items, the keys: + MENU, ↑↓ ±1 HOUR and 9 PRLCD (SNAP on the C47).

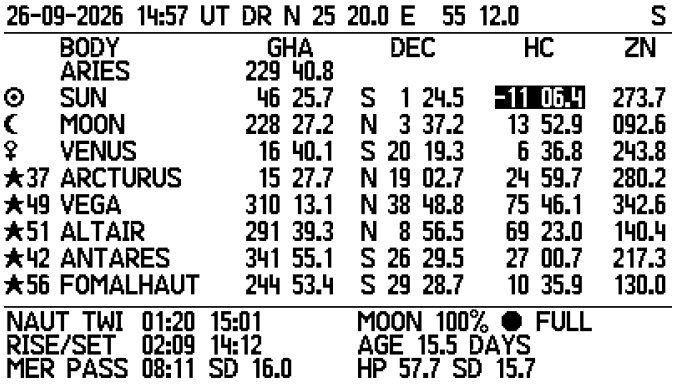


5. Keys

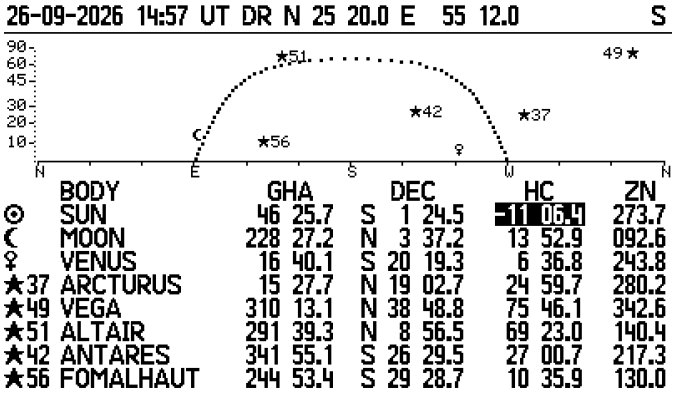
Key	On the menu	On a view
1 – 6	the item is highlighted and its view is drawn (6: INFO)	—
9	SNAP: here PRLCD, the screen to the printer (a screenshot)	PRLCD
+	—	back to the menu
▲ / ▼	one hour later / earlier	the same view one hour later / earlier
+/-	UT or local time (LT = UT + TZ): the menu drawn again	the same: the view drawn again
0	end of NAV: your registers, SIZE, X Y Z T and LASTX back, the normal screen	—
R/S, EXIT	stop the program	stop the program

6. The views

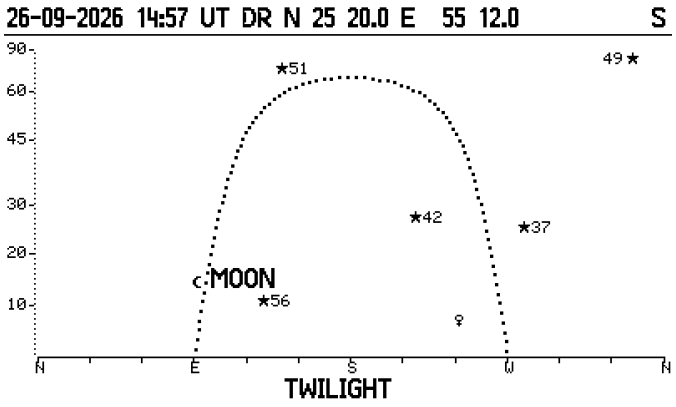
26 Sep 2026 14:57 UT, 25° 20′ N 055° 12′ E, as drawn by Free42.



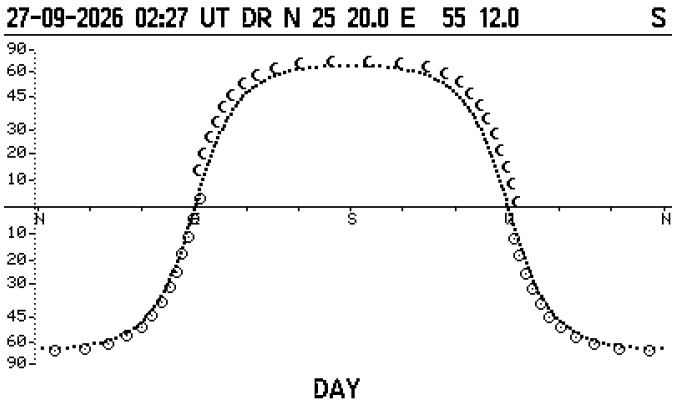
1 ALMANAC: GHA, Dec, Hc, Zn of 10 bodies; twilight, rise/set, mer. pass, Moon phase, HP, SD. A negative Hc is white on black.



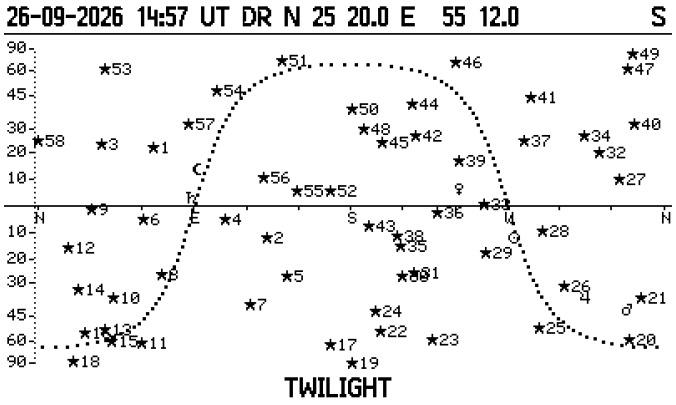
2 SPLIT: chart on top (Hc up, Zn across, the celestial equator dotted), the bodies below down to the bottom.



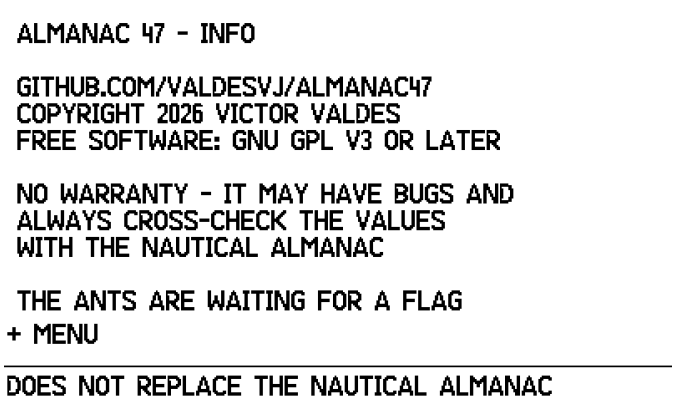
3 SKY: full-screen chart; every 2 seconds the name of the next body is shown next to it; DAY / TWILIGHT / NIGHT at the bottom.



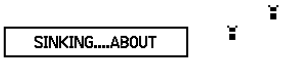
4 ANIM: the Sun and the Moon over 12 hours (24 frames, 1 s each).



5 ALLSKY: the whole sky, over and under the horizon; DAY / TWILIGHT / NIGHT.



6 INFO: repository, licence, no warranty, cross-check (the warning is on the menu and INFO only).



The SINKING....ABOUT box while the sky is computed at the start, with the ants (flag 97, below).

7. How the screen appears

Free42 sends every drawing step to the LCD at once, so each view is drawn in front of you (the stars of SKY appear one by one); the SINKING box shows only while the sky is computed at the start. The C47 shows its screen only at a PAUSE, a key press or the end: there each view appears complete.

NAVFULL and **NAVLITTLE** do it the C47 way with the DM42 variable RefLCD: 0 STO "RefLCD" (no LCD update while NAV computes), −1 STO "RefLCD" (one update) where the C47 program shows its screen, and 7 (normal) when NAV ends. If you stop it with R/S or EXIT and the screen stays frozen, key **7 STO "RefLCD"**.

8. The ants

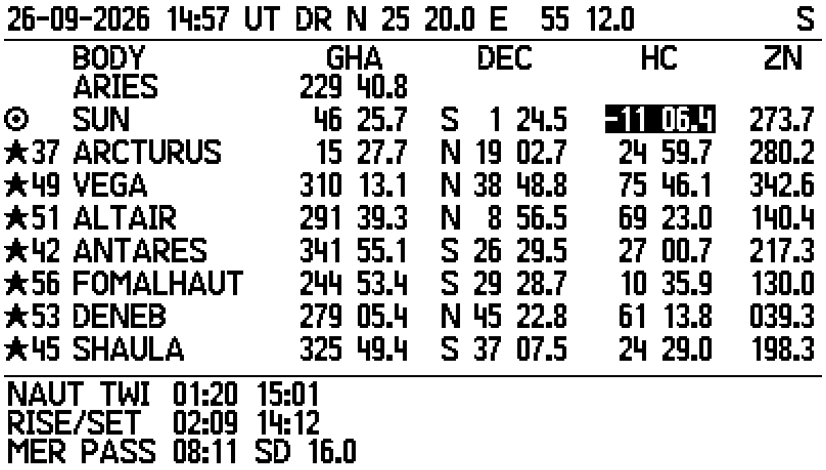
The ants live in NAV as on the C47, but they wait for **flag 97** (the HP-42S flag 47 is a system flag): SF 97 and press a view number, + or an arrow.

9. What is different inside (the screens are the same)

- AGRAPH draws the ALPHA register (8-pixel columns, bit 0 at the top): the fonts are strings of column bytes; the C47 drawing modes (OR, set, clear, XOR = GRMOD 0–3) are the HP-42S AGRAPH flags 34 and 35.
- PIXEL goes through a routine that converts the C47 coordinates (row 0 at the bottom) to the DM42 ones (row 1 at the top).
- Keys (GETKEY / GETKEYA) are translated to the C47 key codes; pauses use TIME; strings use XSTR, APPEND, HEAD; the date is computed (Free42 has no C47 date functions).
- **Your registers are kept**: NAV stores REGS (all your registers, and so your SIZE) in the variable NBAK, runs with SIZE 100 (NAVLITTLE: 38: its values and its most used numbers), and stores NBAK back into REGS when you leave (0; NAVLITTLE: +). If you stop NAV with R/S or EXIT: RCL "NBAK" STO "REGS" gives them back by hand.
- v2.0.0 was faster and smaller than v1.1 (NAVFULL.raw 28.3 KB, was 35.5 KB) with the same screens: Horner, n-vectors, counted loops on ISG, the registers renumbered. v2.1.0 (Supercharger): the tables by GETM and DOT, Hc and Zn straight from the body's direction vector, the Moon's series arguments once, the view header and compass as shared routines, no unused code, the most used numbers in registers, repeated steps as subroutines: NAVFULL.raw 23.3 KB, NAVLITTLE.raw 7.8 KB, MOON47.raw 6.1 KB, the same screens (docs/OPTIMIZATIONS.md). The ↑ ↓ ± of the menu are drawn by the font program from the C47 standard font.
- Built with tools/build_free42.py from the C47 programs; tested with tools/f42 (the Free42 core with the DM42 graphics, binary arithmetic). NAVFULL tested on a DM42n with the stock firmware by the author.

10. NAVLITTLE: Sun and stars

The ALMANAC screen of NAVFULL (the header line with a line under it, the same fonts) with the Sun and the 58 navigation stars (no Moon or planet rows), no box and no ants: for a calculator with little free memory. No Moon: under the table only the Sun's twilight, rise / set and meridian passage; the same screen as the C47 NAVLITTLE. The Sun and star values are the same as in NAVFULL. If a TBL has been run (section 11), the Sun and GHA Aries come from the tables (T on the screen).



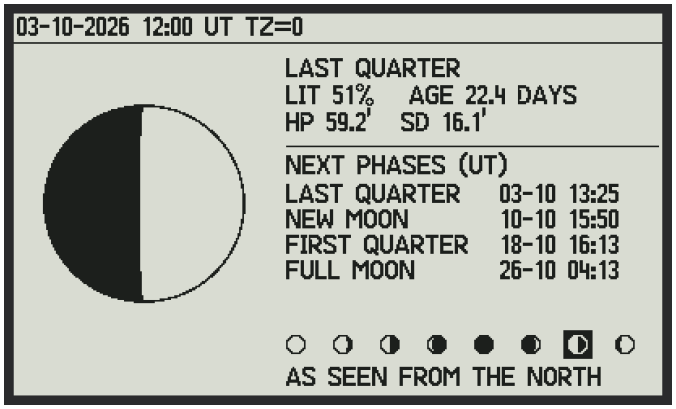
- Load NAVINIT_LITTLE.raw, **XEQ "INIT"** (MATRICES READY: SUN STARS 2000-2050), delete INIT; load NAVLITTLE.raw, **XEQ "NAV"**. If INIT is still loaded and has not run, the first NAV runs it (flag 81).
- NAV asks DATE, UTC, LAT, LON as above and goes straight to the ALMANAC screen: ▲ / ▼ one hour later / earlier (the screen stays while it computes), +/- UT or LT, + ends (GrMod 0; your registers, X Y Z T and LASTX back).
- Programs: NAVLITTLE 8.7 KB, NAVINIT_LITTLE 8 KB (.raw files), matrices 667 numbers.
- Tested in the Free42 core against the same screen in the C47 simulator, pixel by pixel (tests/test_f42_little.py); not yet on a real calculator.

11. Almanac tables: TBL_1, TBL_5 and TBL_50

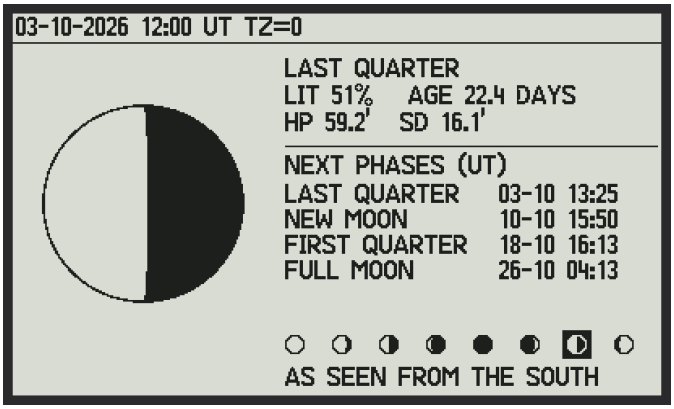
- Optional: Chebyshev coefficients fitted to JPL DE421. **TBL_1**: 1 Oct 2026 – 30 Sep 2027 (8,441 numbers, file 97 KB). **TBL_5**: 1 Oct 2026 – 30 Sep 2031 (41,882 numbers, file 481 KB).
- Load ONE, **XEQ "TBL"** once: it builds the matrices TSU, TVE, TMA, TJU, TSA, TMO, TAR and sets flag 10. Then delete TBL; the matrices stay (about 135 KB for 1 year, 670 KB for 5 years).
- NAVFULL and NAVLITTLE then take the Sun, the Moon, the planets and GHA Aries from the tables inside their period (NAVLITTLE: the Sun and Aries), and the screens show **T** instead of S. Outside the period, or after **CF 10**, the series are used. The values agree with the series to 0.1′.
- The tables need more memory than the C47 has: they are meant for Free42 / Plus42 on a PC or phone, or a DM42n with enough free memory.
- **TBL_50**: 1 Jan 2000 – 31 Dec 2050 (file 4.9 MB, about 6.8 MB of matrices): Free42 / Plus42 on a PC only. Not in the package: download TBL_50.raw from the release tbl50-2000-2050 (github.com/valdesvj/almanac47/releases/tag/tbl50-2000-2050) or make it with tools/almanac/README.md.

12. MOON47: the Moon phase

MOON47 is a separate program, the Moon's phase page without NAV, INIT or the tables: build/free42/MOON47.raw (about 6.5 KB). Load it and **XEQ "MOON47"**: it asks nothing and takes the date and time from the DM42's clock (any date format: YMD, DMY or MDY).



3 Oct 2026 12:00 UT, from the north



after 0: from the south

- **The page:** the Moon as a disc (the lit part filled), the phase name, % lit, age in days, HP and SD, the next new Moon, first quarter, full Moon and last quarter (date and time UT), and the eight phase symbols with today's one inverted. The phase does not depend on where you are; the latitude only turns the picture.
- **Keys:** 0 turns the picture to the view from the south and back; +/- shows the times in UT or in local time (LT = UT + TZ, kept in TZLT); any other key ends.
- **The clock is local time.** If it is not on UT, store your offset once in the variable TZ, in hours: **4 STO "TZ"** for UT+4. Without TZ the clock is taken as UT: the phase times stay right, the age and % lit are then a few hours off. The top line shows the clock: the time UT and TZ=0, or with TZ the local time LT and your offset (TZ=+4, TZ=-5, TZ=+5:30).
- The same screen as MOON47 on the C47, pixel for pixel (tests/test_moon47_f42.py in the Free42 core). It sets SIZE 100 and uses R00–R61; GrMod 3 while it runs, RefLCD so the page appears at once.
- Without the variable TZ the clock is taken as UT (flag 25 ignores the missing TZ).
- **Accuracy** against the full series of NAV, 2000–2050: phase times and age within 4 minutes (most within 1), HP 0.03′, SD 0.01′, % lit 0.03 %: the 20 largest terms of the Moon (Meeus, ch. 47), the same formulas as MOON47 on the C47 (python/moon47.py; tests/test_moon47.py).